IQRASEC ACADEMY

Active Reconnaissance Report

WAPT Phase 1

Task 2 Reconnaissance and Enumeration

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Learning platform

TryHackMe

Description:

This report covers direct information gathering using the Active Reconnaissance room's browser and network exercises. It explains the techniques and provides spaces for practical observations and supporting screenshots.

1. Objective

To inspect a web application through browser tools, measure lab-host reachability, examine network paths and retrieve service banners. The aim is to interpret live responses accurately and distinguish observed information from assumptions about systems or vulnerabilities.

2. Concept and Understanding

2.1 Active Reconnaissance

Active reconnaissance sends traffic directly to a target, unlike reviewing existing third-party information. Browsing a target page, sending ICMP probes and connecting to a service are direct interactions that may be recorded in server, firewall or monitoring logs. The practical activities in this report use the supplied TryHackMe exercises.

2.2 Browser Inspection

Developer Tools show requests, responses and files loaded by a page. The Network panel exposes methods, status codes and headers. Sources or Debugger allows examination of JavaScript. Source content can reveal application structure or data that is not immediately visible in the rendered page. These observations are clues rather than proof of a security weakness.

2.3 Ping and Reachability

For IPv4, ping sends ICMP Echo Requests and receives Echo Replies when the host and network allow them. The output includes response timing, received TTL, packet counts and packet loss. No reply may result from filtering, routing problems or an unavailable host. TTL can suggest an initial configuration, but it is not a reliable operating-system identification on its own.

2.4 Traceroute and Network Paths

Traceroute uses probes with increasing TTL values to reveal responding hops. A router may return an ICMP Time Exceeded message when a probe expires. An asterisk indicates that a probe did not receive a response within its wait period. It does not by itself prove that ordinary traffic is blocked. Routes can change, and responding routers may belong to third-party networks.

2.5 Telnet Netcat and Banners

Telnet and Netcat can connect to TCP services and display responses. A banner or HTTP Server header may identify software and a version, but it may also be hidden, outdated or customized. Plain Telnet and ordinary nc connections do not provide TLS encryption. The port number suggests a conventional service but does not establish what is actually running.

3. Practical Work

3.1 Lab Setup and Target Identification

Use an AttackBox or Kali connected to the TryHackMe network. Start the machine supplied in Task 3 for the ping, traceroute and Telnet exercises. Task 6 supplies a separate machine for its Netcat exercise. Use the IP currently displayed for each task.

3.2 Web Browser Exercise

Open the website linked in Task 2 and launch Developer Tools with Ctrl+Shift+I. Inspect the loaded source files and locate the information needed to determine the total number of questions. Capture the relevant source or data and explain how it supports the count.

3.3 Ping Exercise

On Linux, -c limits the number of requests. The room asks how many replies are received; use your actual summary rather than assuming all ten succeed. The -s option sets the ICMP payload size when exploring the related concept question.

3.4 Traceroute Exercise

traceroute 192.168.200.5

Use the Task 3 machine's IP and record the responding hop lines and whether the destination was reached. The Windows equivalent is tracert. Distinguish the destination line from intermediate routers when counting them. If the trace is incomplete, state that the full path was not established.

3.5 HTTP Banner Retrieval with Telnet

telnet 192.168.200.5 80

Connect to the Task 3 machine. After the connection opens, type the following HTTP/1.1 request, replacing MACHINE_IP in Host with the same lab IP. Finish the headers with an empty line by pressing Enter twice after the Host line.

GET / HTTP/1.1
Host: MACHINE_IP

Record the response status and Server header, if present. The sample server shown in the room is illustrative; your lab response may identify different software. This is a plaintext HTTP exercise on port 80, not a TLS connection.

3.6 FTP Banner Retrieval with Netcat

nc 192.168.200.5 21

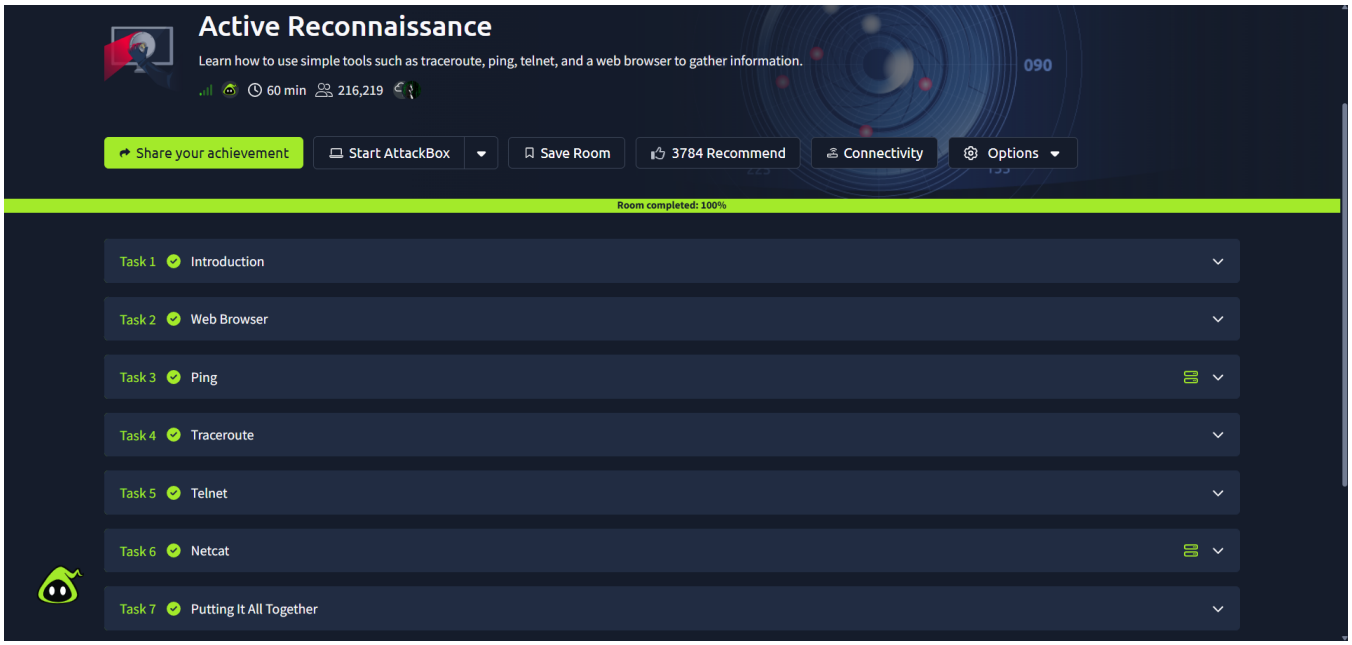
Replace MACHINE_IP with the separate Task 6 machine's IP. Read the initial service banner and record the software version requested by the exercise. A banner is sufficient for this activity; no FTP login or file transfer is required. Stop the connection with Ctrl+C after recording the output.

Netcat can also support client-server communication in listen mode. Command options vary by implementation; check nc -h before any optional practice. Record additional activities only if performed.

4. Tools Used

Tool	Purpose
Browser Developer Tools	Inspect application requests and loaded source files.
ping	Send a bounded set of ICMP reachability probes.
traceroute	Observe responding hops toward the lab host.
telnet	Send an HTTP request and inspect the port 80 response.
Netcat nc	Connect to the Task 6 FTP service and read its banner.
AttackBox or Kali terminal	Run the lab commands using the appropriate task IP.

5. Evidence



7. Key Learnings

Combining browser inspection, reachability probes, path tracing and service responses gives a clearer view of the lab target. Each technique has limits: silence is not proof that a host is offline, TTL is not conclusive OS identification, routes can vary and banners may be incomplete. Results should remain tied to the correct machine, date and evidence.

8. References

TryHackMe. Active Reconnaissance. https://tryhackme.com/room/activerecon

Chrome for Developers. Network panel. https://developer.chrome.com/docs/devtools/network/overview

Nmap Project. Ncat basic usage. https://nmap.org/ncat/guide/ncat-usage.html

iputils. ping manual. https://manpages.debian.org/bookworm/iputils-ping/ping.8.en.html